

Technology Stack

Date	27 June 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Bootstrap, JavaScript	Used to create a responsive and user-friendly web interface for collecting applicant details and displaying prediction results.

2	Backend / Server-Side)	Python, Flask	Flask provides a lightweight framework for integrating the machine learning model with the web application and handling prediction requests efficiently.
3	Database / Data Storage	CSV Dataset, Pickle (.pkl) Model Storage	CSV files are used for training data storage, while Pickle is used to save and load the trained machine learning model for real-time predictions.
4	Cloud / Hosting / Deployment	IBM Watson Machine Learning, IBM Cloud	Enables cloud deployment of the trained model and supports scalable real-time credit card approval predictions.
5	Version Control & CI/CD	GitHub	Used for source code management, version tracking, team collaboration, and project maintenance.
6	Third-Party APIs / Other Tools	Scikit-Learn, XGBoost, NumPy, Pandas, Matplotlib, Seaborn	Scikit-Learn and XGBoost are used for model training and prediction. NumPy and Pandas support data preprocessing, while Matplotlib and Seaborn are used for data visualization and analysis.

Additional Technologies Used

Category	Technology
Programming Language	Python
Machine Learning Algorithms	Logistic Regression, Decision Tree, Random Forest, XGBoost
Data Analysis	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Model Deployment	Flask, IBM Watson Machine Learning
Version Control	GitHub